
Screening on the Wrong Null: When a No-Content Hypothesis Does Not Imply a Zero Statistic

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Abstract

Searching a large candidate space for interesting patterns requires a null distribution for whatever statistic the search screens on, and that null is almost always assumed rather than measured. We study a failure mode that survives every multiplicity correction: a statistic whose value under the *no-content* hypothesis is systematically non-zero. Screening on such a statistic yields discoveries that are real rather than spurious, so raising the threshold for the number of trials does not remove them. Data mining has met this before — the confidence of an association rule between independent items equals the consequent’s marginal frequency, so confidence-based screening harvests common consequents, and lift exists to re-centre the statistic — and the same structure appears in accuracy against a uniform baseline, in recommendation metrics with sampled negatives, and in any improvement-over-baseline score whose baseline is itself estimated.

Empirical asset pricing supplies the one candidate space we know of that is null by construction and large enough to measure the effect: 19,380 long-short equity strategies built from the letters of ticker symbols. Their raw mean-return t -statistics are close to standard normal (SD 1.026); their factor-model alpha t -statistics are not (SD 1.405), and 5.2 percent cross $|t| > 3$ against the 0.27 percent a standard normal implies. Least squares makes the reason exact: $\hat{\alpha} = \bar{r} - \hat{\beta}'\bar{f}$, so a strategy that bears risk and earns nothing has a true alpha of $-\beta'\mu_f$. Sorting the population by exposure moves $\Pr(|t_\alpha| > 3)$ from 0.3 to 24.6 percent while the median standard error moves by a factor of 1.19, a zero-exposure placebo returns 1.001, and a second no-content population is widened by the same factor.

Running the corrections on this population, where every rejection is false by construction, Bonferroni at $M = 19,380$ rejects 0 on the raw mean return and 69 on the alpha; substituting the measured null for the standard normal takes that to 0. The multiplicity arithmetic is not what fails, the null it is given is, and the remedy is the one data mining already adopted: re-centre or recalibrate the statistic on the candidate population being searched. We release `bzoo`, a tested implementation of twelve corrections behind one interface. Code: <https://github.com/0xpinara/benchmark-zoo>.

1 Introduction

Any search over a large candidate space needs a null distribution for the statistic it screens on. The search reports whatever scores highest, so the question “is this real” becomes “is this larger than the best of N draws from the null”, and the answer depends entirely on what that null is. In practice it is assumed rather than measured: a standard normal for a t -statistic, a uniform baseline for an

accuracy, chance for a hit rate. Multiplicity corrections — Bonferroni, Holm, false discovery rate control, resampling-based family-wise procedures — all take that null as an input and correct only for N .

This paper is about a failure mode that no amount of correcting for N can touch: the statistic being screened on has a systematically non-zero value under the hypothesis of *no content*. Data mining has already met this problem and solved it in one instance. The confidence of an association rule $X \Rightarrow Y$ is $P(Y | X)$, so for a rule with no association at all the confidence is $P(Y)$, not zero. Screening rules on confidence therefore harvests frequent consequents rather than genuine associations, and the field’s response was not a stricter threshold but a re-centred statistic: lift, or interest, $P(Y | X)/P(Y)$, which equals one exactly when there is no association [Brin et al., 1997]. The threshold was never the problem. The statistic was.

We show that empirical asset pricing has the same problem, has not noticed it, and pays a measurable price. The reason to study it there rather than anywhere else is evidential: asset pricing supplies the only candidate space we know of that is large and *null by construction*, so the false-discovery rate can be measured against ground truth rather than estimated. Chen and Dim [2025] build roughly 19,000 long–short equity strategies from the letters of ticker symbols. A strategy that buys firms whose ticker begins with a letter in one range and sells firms in another has no economic content by design, so every discovery made on this population is by definition false, and a correction’s promise can be checked rather than assumed.

The statistic the field screens on is a factor-model alpha, and least squares makes its behaviour under the no-content hypothesis exact: $\hat{\alpha} = \bar{r} - \hat{\beta}' \bar{f}$, so a strategy with zero expected return and non-zero factor loadings has a true alpha of $-\beta' \mu_f$. Alpha is confidence, not lift. On the ticker population the raw mean-return t -statistic behaves as the textbook says, with standard deviation 1.026, while the five-factor alpha t -statistic has standard deviation 1.405, and 5.2 percent of empty strategies clear the $|t| > 3$ bar that Harvey et al. [2016] propose for a new factor — 19 times the nominal rate. Running the corrections on the population directly, Bonferroni at the full family size rejects 0 strategies on the mean return, exactly the family-wise control it promises, and 69 on the alpha. Keeping the correction and substituting the measured null for the standard normal takes that to 0: the arithmetic is sound and the null it is handed is not.

One boundary on the claim belongs here rather than in the limitations. Alpha is doing exactly what alpha is defined to do — a strategy that earns nothing while bearing priced risk genuinely does underperform the model, and its negative alpha is a correct statement about it. The claim is about alpha used as a *screening device* for finding predictive signals, where the quantity screened on responds to exposure the signal has no content to justify. Nothing here says the five-factor model is wrong or that risk adjustment should be abandoned, any more than lift says confidence is wrong. It says that the statistic answering “does this beat the benchmark” is not the statistic answering “is there anything here”, and that the field’s threshold is set on the first while the question is usually the second.

We contribute: (i) a characterisation of the failure mode — statistics that are non-zero under the no-content hypothesis — with the association-rule precedent made explicit and three further instances identified (§8); (ii) its measurement on the one candidate space that is null by construction, including the exact variance decomposition the identity implies and four checks that the channel is exposure rather than the regression, replicated on a second no-content population; (iii) the direct test that the assertion in (i) requires — twelve corrections run on a population where every rejection is false — showing that the multiplicity arithmetic is sound and the assumed null is not, together with the split of the inflation into a fixed component a calibration removes and a growing component nothing removes (§4.8); and (iv) *bzoo*, a tested implementation of twelve corrections behind one interface, each with a unit test against a published worked example or a published property of the procedure (§9). The corrections themselves are decades old and none is new here; what is new is that the null they need is measured on a population where the measurement can be checked.

2 Related work

Pattern discovery and re-centred statistics. The same problem is native to data mining, where searching a large pattern space guarantees significant-looking patterns. Webb [2007] and Hämmäläinen and Webb [2019] set out the framework; Terada et al. [2013] and Llinares-López et al. [2015]

make permutation-based family-wise control tractable over very large pattern sets. The problem this paper describes has the same shape: a large space of candidate signals, searched with a statistic whose null the searcher has assumed rather than measured.

The part of this literature we lean on hardest is the one that fixed a statistic rather than a threshold. Brin et al. [1997] observe that the confidence of a rule with no association is the marginal frequency of its consequent, and introduce lift to re-centre it at one; the same move appears whenever a measure of association is normalised by what independence would give. Our remedy in §7 is the asset-pricing version of that move, arrived at independently by a literature that has spent two decades raising thresholds instead.

Multiplicity corrections. We implement the classical family: Bonferroni [1936] and Šidák [1967], the step-down procedure of Holm [1979], the false discovery rate of Benjamini and Hochberg [1995] and its arbitrary-dependence version [Benjamini and Yekutieli, 2001], and the q -values of Storey [2002] in the bootstrap form of Storey et al. [2004]. The resampling family learns dependence from the data rather than assuming it: White [2000], Hansen [2005], Romano and Wolf [2005] and the permutation procedures of Westfall and Young [1993]. Efron [2004] makes the point this paper turns on — that the null in a large-scale testing problem is an empirical object — in the microarray setting; the finance and machine learning literatures have generally not followed.

The application domain: the factor zoo. Harvey et al. [2016] apply Bonferroni, Holm and Benjamini–Hochberg–Yekutieli to published return predictors and argue for a t -statistic bar near 3.0; Harvey and Liu [2015] turn an adjusted p -value into a haircut on a reported Sharpe ratio; Bailey and López de Prado [2014] deflate a Sharpe ratio by the expected best of N trials. Chen and Zimmermann [2022] rebuild 212 published predictors from source and publish the portfolios, and Jensen et al. [2023] argue the replication record is better than the multiple-testing literature implies. Yan and Zheng [2017] and Chen and Dim [2025] mine large strategy populations; the ticker-symbol strategies of the latter are the known-null population we calibrate on, and Chen and Dim already observe that their t -statistics sit close to a normal null. Harvey and Liu [2020] give the current statement of the false-discovery problem in this literature.

Nonsense predictors, and how this differs. Novy-Marx [2014] is the closest published neighbour: he shows that politics, the weather, sunspots and the stars “predict” anomaly performance, and the lesson drawn is about data mining — search hard enough over unrelated series and something will fit. Our claim is not that one. The ticker strategies here are not selected for looking good; the entire population is reported, its hit rate is measured against what the null promises, and the inflation is traced to an identity rather than to a search. A nonsense predictor found by mining is a chance result that a multiplicity correction is designed to catch. An empty-but-exposed strategy is not a chance result, and Table 6 shows the correction does not catch it. Lo and MacKinlay [1990] make the data-snooping point for asset-pricing tests generally, Ferson et al. [2003] treat spurious regression when the regressor is persistent, and Fama and French [2010] bootstrap a null with zero alpha injected by construction — methodologically the closest thing to our sign-flip permutation, in the mutual-fund setting.

Our contribution on this side is to quantify the departure, to extend it to the statistics papers actually report, and to show where it stops being true.

3 Data

The known-null population. Chen and Dim [2025] distribute monthly returns for two mined populations of long–short strategies. We use both, for opposite purposes. The *ticker* population contains 19,380 signals: stocks are sorted by one letter of their ticker symbol, split into twenty groups, and the strategy is long two groups and short two others, over all $\binom{20}{4}$ choices and all four letter positions. The *past-return* population contains 19,402 signals built from lagged own returns and their higher moments. Each signal appears in an equal-weighted and a value-weighted version, which are two strategies, so the 19,380 ticker *signals* are 38,760 strategies and we keep the two weightings in separate panels throughout. The ticker sample is 720 months, January 1963 to December 2022, for the first three letter positions; the fourth letter exists only for tickers of four characters or more and has 600 months, which is why 14,535 rather than 19,380 signals have a

Table 1: Mined strategy populations from Chen and Dim, after the classification rules of Section 3. Ticker strategies are null by construction and are used to calibrate; past-return strategies contain known effects and are used to check power.

Population	Signals	Role	Composition
ticker	19 380	known null	L1: 4845, L2: 4845, L3: 4845, L4: 4845
pastret	19 402	known signal	other: 19317, longrun: 80, momentum: 4, recent: 1

complete history. The past-return sample starts a year earlier and runs to the same month. Every statistic below is computed on each strategy’s own available months, with a sixty-month minimum.

The ticker population is the known null and the past-return population is the known-signal set used to check power. Getting a past-return strategy into the null set would inflate the estimated null and make every threshold in this paper too strict, so the classification is written out as ten explicit rules, unit tested against hand-labelled names, and reproduced in `src/bzoo/finance/partition.py`. The upstream files are separate, which removes most of the risk, but their signal-id spaces both start at zero, so any pooled analysis has to namespace them.

The indices in a past-return signal name are *quarters* ordered oldest first, not month lags, and reading them the other way inverts every economic interpretation. Under the correct reading the known-signal set behaves as the literature says with no tuning — twelve-month momentum has $t = 5.60$ equal-weighted, the most recent quarter alone is negative, and four-to-five-year returns are negative — which is the power check. The rules and the failed alternative reading are in `src/bzoo/finance/partition.py`.

Published predictors. Chen and Zimmermann [2022] publish long–short portfolio returns for 212 predictors documented from their original papers, with each predictor’s original sample window. 208 have at least sixty months in that window. Recomputing their t -statistics from the distributed returns and comparing with the t -statistics recorded from the original papers gives a correlation of 0.75; the median recomputed $|t|$ is 3.20, and 88 percent of them exceed 1.96. The release we use documents 212 predictors, 114 placebos and 5 dropped signals; the `openassetpricing` client exposes portfolio returns for the predictors only, in seven weighting and filter variants, and has no placebo path. The placebo *characteristics* are distributed at the firm level, so a placebo portfolio population could be built with the sorting code this package already has. We have not built it, and it would be the right third no-content population, because a characteristic that was examined and not claimed models researcher behaviour in a way a ticker letter does not.

Licences and scope. The mined strategy returns and the OSAP portfolios are distributed publicly by their authors; we redistribute derived statistics only, never the returns. The factor returns are Kenneth French’s. Our code is MIT.

4 Measuring the null

4.1 What can go wrong, kept apart

Three things can make a theoretical null wrong, and they have different consequences, so we measure them separately. **Scale:** if the null t -statistics have standard deviation 1.4 rather than 1.0, every nominal threshold understates the true one by 40 percent. **Tail shape:** even at the right variance, a heavier tail puts more mass past any cutoff, and the cutoff is where a correction operates. **Dependence:** this leaves the marginal distribution untouched and changes the distribution of the *maximum*, which is the object every family-wise correction actually uses. A population can have an exactly standard normal marginal null and still make Bonferroni wrong, in either direction. Reporting only a variance inflation factor, as is common, cannot distinguish these.

Table 2: The measured null. Cross-sectional distribution of four statistics over the 19,380 ticker-symbol strategies, which have no economic content by construction. The theoretical null has SD 1.00, kurtosis 3.00, $\Pr(|t| > 1.96) = 0.050$ and $\Pr(|t| > 3) = 0.0027$. The last column is the cutoff that gives a 5 percent per-test false positive rate against the measured null, with no multiplicity correction.

Statistic	Weighting	SD	MAD-SD	Kurtosis	$\Pr(t > 1.96)$	$\Pr(t > 3)$	5% cutoff
CAPM alpha t	EW	0.883	0.905	2.73	0.0446	0.00010	1.92
Five-factor alpha t	EW	1.150	1.151	2.97	0.0915	0.00820	2.27
Mean return t	EW	0.889	0.899	2.81	0.0300	0.00005	1.77
Mean return t , Newey–West	EW	0.886	0.888	2.86	0.0307	0.00005	1.78
CAPM alpha t	VW	1.070	1.066	2.96	0.0724	0.00531	2.14
Five-factor alpha t	VW	1.405	1.342	3.30	0.1674	0.05232	3.04
Mean return t	VW	1.026	0.990	3.11	0.0628	0.00470	2.08
Mean return t , Newey–West	VW	1.054	1.000	3.27	0.0720	0.00738	2.16

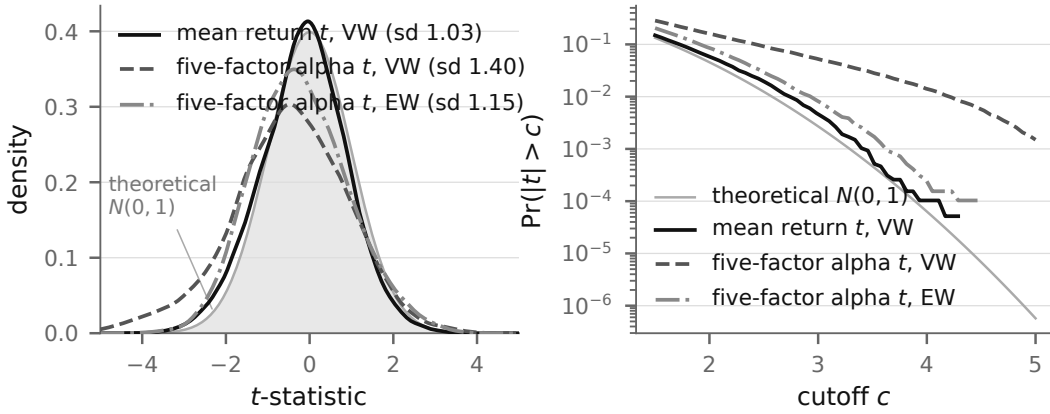


Figure 1: The measured null against the theoretical one, on a population that is null by construction. Left: densities, with the standard normal shaded. Right: the upper tail. The mean-return t -statistic is close to nominal; the five-factor alpha t -statistic, which is what published papers report, is not, and the gap widens with the cutoff. The empirical curves flatten below about 5×10^{-5} , which is the resolution floor of 19,380 strategies.

4.2 Scale and shape

Table 2 is the measurement. Figure 1 is the same thing as a picture, with the tail on a log scale because that is where a correction lives and a linear density shows nothing there.

Two findings. First, for the raw mean-return t -statistic the theoretical null is a good approximation, and equal-weighted it is if anything slightly conservative: standard deviation 0.889 rather than 1.00. The construction imposes an approximate zero-sum constraint across strategies within a letter position — every group appears equally often on the long and the short side — which shrinks the cross-sectional spread. Value-weighted the figure is 1.026, essentially nominal. This confirms Chen and Dim’s qualitative statement and puts a number on it.

Second, the statistic published papers report behaves differently. The five-factor alpha t -statistic has cross-sectional standard deviation 1.150 equal-weighted and 1.405 value-weighted; against the nominal 5 percent cutoff, 16.7 percent of no-content strategies are significant, and at $|t| > 3$ the figure is 5.2 percent against a nominal 0.27 percent. Kurtosis stays near 3 throughout, so this is a scale effect and not a tail-shape effect — which is what makes the rescaling of §4.8 sufficient, since a scale effect is correctable by one number and a tail effect is not. Generalised Pareto fits to the upper tail and their threshold-stability diagnostics are in `data/results/empirical_null.json`. §4.4 explains where that width comes from, and it is not sampling noise.

4.3 Is the population really null in raw returns?

Null in this subsection means null in raw returns, which is the premise everything after it rests on. Section 4.4 shows the same population is emphatically *not* null in factor-model alphas, and that is the paper’s point; the two statements are compatible and the identity $\hat{\alpha} = \bar{r} - \hat{\beta}' \bar{f}$ is why.

A referee’s first objection is that ticker letters are not neutral: alphabetical position is documented to affect turnover and valuation [Itzkowitz et al., 2016]. We test 18 subgroups — the four letter positions, three alphabetical regions of the long leg, and extreme against interior sorts, in both weightings — for a systematic non-zero mean, aggregating each subgroup into one monthly series first and using a stationary block bootstrap over months, because strategies within a subgroup share months and are strongly dependent. Every interval contains zero, and no subgroup is flagged at any multiplicity correction we apply. Dropping every strategy whose sort touches the extreme ends of the alphabet leaves the calibration unchanged (Table 8). The aggregate mean of the whole ticker population is 0.0073 percent per month, $t = 0.50$.

Recalibrating separately on the four disjoint letter-position subsets gives a standard deviation of the mean-return t -statistic in the range 0.739 to 1.045 equal-weighted, and of the five-factor alpha t -statistic 1.103 to 1.590 value-weighted. The calibration is reproducible in sign and rough magnitude on 4,845 dependent strategies and not to two decimal places, and we report it as a range.

4.4 Where the known-null assumption fails

The wide five-factor null is not a miscalibrated standard error. It is a real non-zero alpha, and the reason is arithmetic. Ordinary least squares gives $\hat{\alpha} = \bar{r} - \hat{\beta}' \bar{f}$ exactly, so a strategy with zero expected return and non-zero factor loadings has a non-zero alpha. Sorting stocks on a ticker letter produces no expected return but does produce factor exposure, because the alphabet is not independent of firm characteristics.

The identity turns into an exact variance decomposition, which is the cleanest way to say how much dispersion risk adjustment adds:

$$\text{Var}(\hat{\alpha}) = \text{Var}(\bar{r}) + \text{Var}(\hat{\beta}' \bar{f}) - 2 \text{Cov}(\bar{r}, \hat{\beta}' \bar{f}). \quad (1)$$

Value-weighted, in squared basis points per month, the three measured terms are $\text{Var}(\bar{r}) = 88.90$, $\text{Var}(\hat{\beta}' \bar{f}) = 52.01$ and $\text{Cov}(\bar{r}, \hat{\beta}' \bar{f}) = -15.44$, which give $\text{Var}(\hat{\alpha}) = 171.81$. The covariance is negative, so the -2Cov term *adds* 30.89 rather than subtracting: strategies with more positive exposure tend to have lower mean returns, and risk adjustment pulls those two apart instead of letting them offset. The alphas end up 1.390 times as dispersed as the mean returns, with 63 percent of the increase coming from the exposure variance and 37 percent from the covariance term. Every term is measured, the identity holds to machine precision, and none of it depends on a distributional assumption.

A negative covariance in a population that is null by construction invites two objections, and working through them says something useful about what the term is. The first is that $\hat{\beta}$ and $\bar{r}$ are estimated on the same months, so the two might be coupled mechanically. They are not: for least squares with an intercept the coupling term is proportional to the sum of the demeaned regressor and is therefore exactly zero, with or without cross-sectional dependence in the residuals. Estimating β on the first half of the sample and forming the exposure on the second, so that the two sides use disjoint data, leaves the covariance negative at -9.5 squared basis points with a 95 percent bootstrap interval of -11.1 to -7.9.

The second objection is that a null population should have no such covariance at all. That is not what the algebra says. Substituting $\bar{r} = \beta'(\bar{f} - \mu_f) + \bar{e}$ gives $\text{Cov}_i(\bar{r}, \beta' \bar{f}) = (\bar{f} - \mu_f)' \Sigma_{\beta} \bar{f}$, a realised quadratic form in the sample factor means whose expectation over samples is $\text{tr}(\Sigma_{\beta} \text{Var}(\bar{f})) > 0$ and which can land either side of zero in any one sample. So the sign here is not evidence that the population fails to be null, and it is not an artefact. It is a sample-specific realisation — the same conclusion the variance split of §4.8 reaches from a different direction, and the reason we do not build any claim on the covariance term’s sign.

Two conventions in this literature deserve a correction here, because both are easy to state and wrong. First, the cross-sectional regression of $\hat{\alpha}$ on $-\hat{\beta}' \bar{f}$ is sometimes said to have a population

slope of exactly one, on the grounds that $\bar{r}$ is centred at zero. Centring fixes the *mean* of $\bar{r}$ and the slope depends on its *covariance*: substituting the identity gives $1 + \text{Cov}(\bar{r}, -\hat{\beta}'\bar{f}) / \text{Var}(\hat{\beta}'\bar{f})$. We measure 1.297 value-weighted (White standard error 0.009) and 0.992 equal-weighted, and the formula reproduces both to machine precision. Second, the correlation between $\hat{\alpha}$ and $-\hat{\beta}'\bar{f}$, 0.72 value-weighted, and the share $\text{Var}(\hat{\beta}'\bar{f}) / \text{Var}(\hat{\alpha})$, 29 percent, are different quantities and do not have to agree: the first squares to 0.51 and the gap between them is exactly the covariance term in (1). We report the decomposition rather than either summary alone.

The standard errors are not doing the work. Across the six specifications of Table 3 — the raw mean return and five factor models — the median $\text{SE}(\hat{\alpha})$ moves from 7.96 to 8.00 basis points, about a percent, while $\text{SD}(\hat{\alpha})$ moves from 9.43 to 13.11. The exact finite-sample cost of estimating β rather than knowing it is $\sqrt{1 + \bar{f}'S_f^{-1}\bar{f}} = 1.046$ for the five-factor model — a squared-Sharpe-ratio quantity that does not shrink with T — and it works in the direction that makes t *smaller*, so the widening we report is a lower bound.

So “no economic content” implies a null raw return and does *not* imply a null alpha.

4.5 Two questions, two nulls

Which question a null is for now becomes the whole issue, because the answer differs and the two are routinely reported as though they were one object.

If the question is *does this strategy have a non-zero alpha* — a test of $\alpha = 0$ — then the marginal cross-sectional spread of the ticker population is the wrong null, for exactly the reason just given: these alphas are genuinely non-zero, so their spread is not the distribution of $\hat{\alpha}$ under $\alpha = 0$ but the distribution of a real quantity. The right null is one that imposes $\alpha = 0$ while leaving the exposures where they are. We construct it by sign-flipping the factor-model residuals in blocks, with the same sign vector across all strategies, which leaves the loadings and the cross-strategy dependence untouched. That gives a 95th percentile of the largest alpha t -statistic of 4.09 value-weighted, against 4.33 with dependence removed, and those are the numbers §5 uses.

If the question is *does this signal have content* — which is what a researcher screening candidate signals is actually asking — then the marginal spread is precisely the right object, and the fact that the alphas are real is what makes it so. It answers: what alpha t -statistics do empty-but-exposed strategies produce? A screener needs to clear that distribution, not a distribution built by assuming away the exposure that the empty strategies actually have. Its 95th percentile, 3.04 value-weighted, is the cutoff we recommend in §7, and it is a screening calibration rather than a hypothesis test.

The two nulls answer different questions and neither substitutes for the other. A signal can have a genuinely non-zero alpha — passing the first test — and still be indistinguishable from a ticker sort — failing the second. That combination is not a contradiction; it is the entire phenomenon this paper describes.

4.6 Four checks on the mechanism, and where the exposure comes from

Across benchmark models. Table 3 and the left panel of Figure 2 run the same strategies over the same months under the raw mean return and five factor models. Value-weighted, $\text{SD}(t_\alpha)$ goes 1.023, 1.070, 1.361, 1.283, 1.405, 1.368 for $K = 0, 1, 3, 4, 5, 6$. It is not monotone in K , and it should not be: what matters is not how many regressors are added but whether the added factor carries a mean return the sort happens to load on. Adding momentum to Fama–French 3 lowers the spread; adding profitability and investment raises it. A count of regressors does not predict the curve and the exposures do.

Within the population. Table 4 sorts the strategies into deciles on $|\hat{\beta}'\bar{f}|$. $\text{SD}(t_\alpha)$ runs from 0.994 in the bottom decile to 1.810 in the top, and $\Pr(|t_\alpha| > 3)$ from 0.3 percent to 24.6 percent. The bottom decile is a placebo drawn from inside the population itself: strategies that happen to carry almost no exposure have an alpha t -statistic of almost exactly the textbook width. Binning on R^2 instead gives a similar picture for a reason that is partly a denominator effect — a higher R^2 means a smaller residual variance and a smaller standard error — which is why the exposure binning is the one we lead with, and why $\text{SD}(\hat{\alpha})$ is reported next to $\text{SD}(t_\alpha)$ in every bin.

Table 3: Dose-response across benchmark models, on the 19,380 ticker strategies. K is the number of factors; every row is the same strategies over the same 714 months, so only the model changes. $SD(\hat{\alpha})$ and the median standard error are in basis points per month. $SE\ infl.$ is $\sqrt{1 + \bar{f}' S_f^{-1} \bar{f}}$, the exact finite-sample cost of estimating β rather than knowing it, which is a Sharpe-ratio quantity and does not shrink with T . The spread of the alpha widens by a third while the standard error moves by about a percent, so the widening is entirely in the numerator.

Model	K	Wt.	SD	MAD-SD	Kurt.	$\Pr(t > 1.96)$	$\Pr(t > 3)$	5% cutoff	$SD(\hat{\alpha})$	Med. SE	SE infl.
Raw mean return	0	VW	1.023	0.985	3.09	0.0618	0.0044	2.07	9.43	7.96	1.0000
CAPM	1	VW	1.070	1.066	2.96	0.0724	0.0053	2.14	9.97	7.95	1.0076
Fama–French 3	3	VW	1.361	1.362	2.93	0.1658	0.0345	2.79	12.31	7.88	1.0173
Carhart 4	4	VW	1.283	1.287	2.96	0.1377	0.0248	2.62	12.06	8.03	1.0398
Fama–French 5	5	VW	1.405	1.342	3.30	0.1674	0.0523	3.04	13.11	8.00	1.0465
FF5 + momentum	6	VW	1.368	1.283	3.42	0.1549	0.0467	2.92	12.97	8.10	1.0628
Raw mean return	0	EW	0.883	0.888	2.82	0.0292	5.160×10^{-5}	1.76	4.58	4.70	1.0000
CAPM	1	EW	0.883	0.905	2.73	0.0446	0.0001	1.92	4.94	4.71	1.0076
Fama–French 3	3	EW	0.909	0.940	2.74	0.0331	5.160×10^{-5}	1.82	4.73	4.68	1.0173
Carhart 4	4	EW	0.941	0.974	2.72	0.0332	0.0002	1.83	5.02	4.78	1.0398
Fama–French 5	5	EW	1.150	1.151	2.97	0.0915	0.0082	2.27	5.99	4.76	1.0465
FF5 + momentum	6	EW	1.213	1.223	2.93	0.1232	0.0143	2.45	6.54	4.83	1.0628

Table 4: Within-population dose-response, value-weighted. Strategies are sorted into deciles by the absolute size of their exposure term $|\hat{\beta}' \bar{f}|$; all quantities except the last two are in basis points per month. Across the deciles the numerator of the t -statistic rises by a factor of two while its denominator rises by a fifth, which is what identifies the mechanism as exposure rather than a standard-error effect. The bottom decile is a within-population placebo: strategies with almost no exposure have an alpha t -statistic of almost exactly the textbook width.

Decile	n	Median $ \hat{\beta}' \bar{f} $	$SD(t_\alpha)$	$SD(\hat{\alpha})$	Median $SE(\hat{\alpha})$	$\Pr(t_\alpha > 3)$	Median R^2
1	1938	0.48	0.994	8.88	7.64	0.0031	0.049
2	1938	1.43	1.013	9.06	7.67	0.0052	0.047
3	1938	2.41	1.025	9.21	7.67	0.0046	0.047
4	1938	3.42	1.084	9.71	7.73	0.0103	0.049
5	1938	4.53	1.126	10.21	7.79	0.0160	0.051
6	1938	5.73	1.250	11.36	7.88	0.0237	0.055
7	1938	7.14	1.309	12.16	8.04	0.0310	0.059
8	1938	8.91	1.486	13.62	8.18	0.0630	0.068
9	1938	11.22	1.646	15.19	8.45	0.1202	0.078
10	1938	15.66	1.810	17.59	9.11	0.2461	0.099

A zero-exposure placebo. Simulating returns with zero mean and zero population exposure, matched to each strategy’s residual volatility, and running the identical five-factor regressions gives $SD(t_\alpha) = 1.001$ under independent normal draws and 0.976 under a bootstrap of the fitted residuals that preserves their dependence and non-normality. The regression machinery does not manufacture the spread; the exposures do.

Persistence, and what does not persist. A spread that is pure estimation noise shrinks like $1/\sqrt{T}$. This one does not: over nested windows the observed dispersion of $\hat{\alpha}$ falls from 19.1 to 13.4 basis points where noise alone would predict 8.5, which is the right panel of Figure 3. That curve is descriptive only, and we do not fit it: nested windows share most of their data, so the points are heavily dependent and any standard error on a fitted intercept would be uninterpretable.

The question the curve raises — is there a component of these alphas that is a fixed property of the strategy rather than of the sample — has a clean answer that needs no functional form. Split the sample into 5 disjoint 120-month blocks and estimate every strategy’s alpha separately in each. Estimation noise is independent across disjoint blocks, so it inflates the within-block cross-sectional variance and contributes nothing to the covariance between two different blocks; the between-block covariance is therefore an unbiased estimate of the persistent variance on its own. Bootstrapping over strategies, that component is 1.62 basis points per month, 95 percent interval 0.63 to 2.28, against a within-block dispersion of 22.5 basis points.

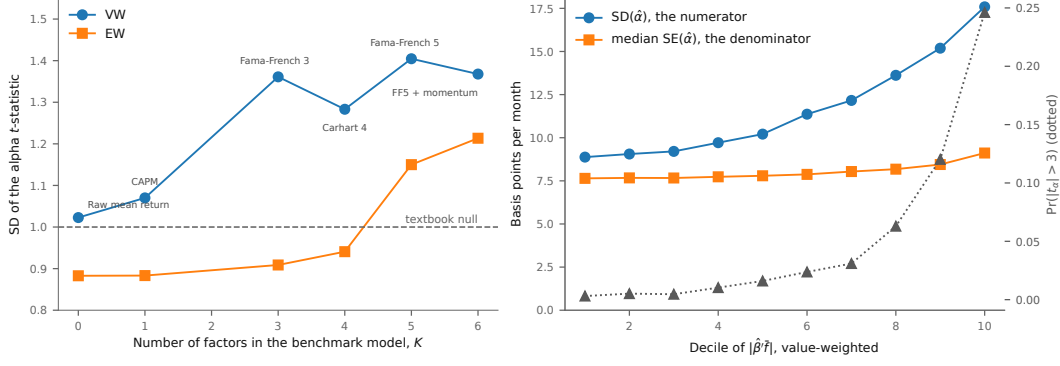


Figure 2: Left: the spread of the alpha t -statistic on a population with no predictive content, under the raw mean return and five factor models. Right: within the value-weighted population, sorted into deciles by exposure. The numerator of the t -statistic rises across the deciles by a factor of 1.98 and the denominator by 1.19, which is what makes this an exposure effect rather than a standard-error effect.

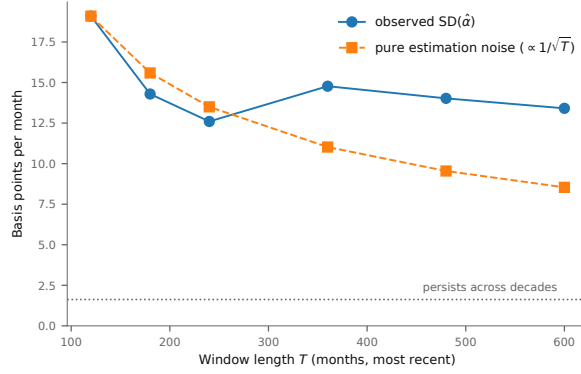


Figure 3: The dispersion of $\hat{\alpha}$ over nested windows of the most recent T months, against what pure estimation noise would give. The dotted line is the component that persists across disjoint decades, estimated separately, and it is far below the gap between the two curves: most of that gap is period-specific exposure rather than a fixed property of the strategy.

So it is different from zero — 0.005 of the bootstrap distribution lies below zero, a one-sided level of the same size — and it is small, only 0.5 percent of the within-block variance. We read that as follows, and it is the more interesting reading. The inflated alphas are real in any given sample, which is what defeats a multiplicity correction. They are largely *not* stable across decades, because the exposure term $\hat{\beta}' \bar{f}$ depends on realised factor premia and those move. A screener who finds one of these has therefore found neither a chance result nor a durable anomaly, but a sample-specific exposure artefact — and both halves of that are bad news, because the first means a stricter threshold will not remove it and the second means it will not replicate.

Where the exposure comes from, and what wins. Decomposing $\hat{\beta}' \bar{f}$ by factor, the investment and profitability terms carry 67 percent of its variance, size 16 percent and the market only 3 percent. The story is not the size effect one might guess from Itzkowitz et al. [2016]: size carries 16 percent and the two factors that dominate are the ones Fama and French added in 2015.

We cannot say why. Letter position and the alphabet regions of the two legs explain $R^2 = 0.105$ of the exposure, so the alphabet does map systematically onto profitability and investment loadings, but weakly, and a weak reduced-form mapping is not a mechanism. The candidate we would test first is industry: firms have historically chosen mnemonic tickers, so ticker letters should cluster by sector, and sector composition would plausibly drive profitability and investment loadings. Testing it needs the industry composition of each long leg, which needs firm-level holdings; Chen and Dim

Table 5: Two no-content populations, built differently, widened the same way. Dispersions of $\bar{r}$ and $\hat{\alpha}$ are in basis points per month; each *Ratio* column is the alpha entry divided by the mean-return entry to its left. The second population’s alpha t -statistics sit near the textbook width in *absolute* terms only because its raw-return null is unusually narrow to begin with: risk adjustment widens its numerator by as much as it widens the ticker population’s. Two independent constructions, one mechanism.

Population	Wt.	n	$SD(\bar{r})$	$SD(t_\alpha)$	Ratio	$SD(\bar{r})$	$SD(\hat{\alpha})$	Ratio
Ticker letters	VW	19 380	1.023	1.405	1.373	9.43	13.11	1.390
Higher-moment, 3–5 yr	VW	210	0.775	1.009	1.301	8.08	11.48	1.420
Ticker letters	EW	19 380	0.883	1.150	1.303	4.58	5.99	1.308
Higher-moment, 3–5 yr	EW	210	0.663	0.887	1.338	5.50	6.57	1.194

[2025] distribute portfolio returns only, so it needs CRSP identifiers and SIC codes that are outside what this paper’s data allows. We flag it as the obvious next step rather than gesturing at it as though it were established. In the top and bottom one percent of alphas the loadings flip sign as the mechanism requires: the strategies with the largest positive alphas are the ones short the priced factors. The single most significant strategy value-weighted is L2_lng_1_8_sht_15_16 — sort on the second letter of the ticker, buy two groups and sell two others. Its three numbers are $\bar{r} = -22.7$, $\hat{\beta}'\bar{f} = 22.7$ and therefore $\hat{\alpha} = -45.5$ basis points per month. Losing -22.7 while carrying enough exposure to have been expected to make 22.7 gives $t_\alpha = -5.68$, which clears Bonferroni at the full trial count, on a raw-return t -statistic of -2.74. 50 percent of the alpha is the exposure, and the strategy has no content at all.

4.7 A second no-content population

The 210 higher-moment past-return strategies built only from returns three to five years old — a different construction, with no documented effect at that horizon — give an independent test. Value-weighted their five-factor alpha t -statistic has standard deviation 1.009, against the ticker population’s 1.405, and it would be easy to read that as the alpha inflation being specific to ticker sorts. It is not, and Table 5 shows why. That population’s *raw-return* null is unusually narrow, 0.775, so the right comparison is not the level but the widening. Risk adjustment multiplies the dispersion of this population’s numerator by 1.42 and the ticker population’s by 1.39. Two populations built from unrelated sorting variables, widened by the same amount. The mechanism replicates.

The transferable lesson is the general form: a population that is null with respect to one statistic need not be null with respect to a transformation of it, and any signal that induces factor exposure without predictive content will produce inflated alpha t -statistics. That is a real false-discovery channel for anyone screening signals on risk-adjusted statistics, and because the alphas are real rather than spurious, no multiplicity correction addresses it.

What would overturn this. Four results would, and we report all four rather than only the ones that came out our way. *One:* a no-content population that carries factor exposure but shows no alpha inflation. Our second population is such a test and it inflates (Table 5). *Two:* a zero-exposure simulation returning $SD(t_\alpha)$ materially above 1. It returns 1.001. *Three:* no relationship between exposure and alpha inflation within a population. The relationship is monotone across all ten deciles and the numerator carries it (Table 4). *Four:* alpha dispersion shrinking exactly like $1/\sqrt{T}$, which would make it estimation noise; it does not, and a disjoint-block estimate puts a small but non-zero persistent component at 1.62 basis points. Two of the checks we ran did not behave as the simple version of the theory predicts — $SD(t_\alpha)$ is not monotone in the number of factors, and the cross-sectional slope of $\hat{\alpha}$ on $-\hat{\beta}'\bar{f}$ is 1.297 rather than 1 — and in both cases the identity predicts the departure exactly, which is reported above.

4.8 What a correction does and does not fix

The claim that a multiplicity correction is no remedy here is directly testable, and Table 6 is the test: apply the corrections to a population where every rejection is by construction a false one. A 5 percent family-wise procedure promises at most a 5 percent chance of *any* rejection across the

Table 6: Multiplicity corrections applied to 19,380 strategies that have no economic content by construction, value-weighted, with the family size set to the full 19,380 and the nominal standard normal used as the null. Every rejection in this table is a false one. A 5 percent family-wise procedure promises at most a 5 percent chance of *any* rejection across the family; on the raw mean return Bonferroni delivers that and rejects nothing, and on the five-factor alpha it rejects 69 times. The multiplicity arithmetic is not what fails — the null it is applied to is.

Correction	Mean return t	Five-factor αt
Uncorrected ($ t > 1.96$)	1 197	3 244
Bonferroni	0	69
Šidák	0	70
Holm	0	69
Benjamini–Hochberg	0	995
Benjamini–Yekutieli	0	292

family. On the raw mean-return t -statistic Bonferroni delivers exactly that and rejects 0 of 19,380. On the five-factor alpha, at the same family size and the same nominal null, it rejects 69. Benjamini–Hochberg rejects 995 against 0. Same strategies, same correction, same M ; the only thing that changed is the statistic.

It is not the multiplicity arithmetic that fails. Substituting the measured null for the standard normal — keeping Bonferroni and keeping $M = 19,380$, and only widening the null from 1.00 to 1.405 — moves the critical value from 4.70 to 6.60 and the false rejections from 69 to 0. The correction works once it is given the right null. What it cannot do is supply that null itself.

Why a fixed rescaling is enough, and where it stops. This deserves an explanation rather than a lucky observation, and the identity supplies one. For a strategy with no content the true alpha is $-\beta'\mu_f$, using population premia, while the estimate is $\hat{\alpha} = -\beta'\mu_f - \beta'(\bar{f} - \mu_f) + \bar{e} + O(1/T)$. The two middle terms behave completely differently in t -units. The first is a fixed quantity divided by a standard error that shrinks like $1/\sqrt{T}$, so it contributes a variance growing like T . The second involves $\bar{f} - \mu_f$, which converges at exactly the rate the standard error shrinks, so it contributes a variance that does not grow at all. Writing $\text{Var}(t_\alpha) = 1 + c_1 + c_2T$, the persistent piece is c_2T and the sample-specific piece is c_1 .

We can separate them, because the disjoint-block estimate of §4.6 measures $\text{Var}(\beta'\mu_f)$ directly. Value-weighted, $c_1 = 0.73$ and $c_2T = 0.027$ at $T = 714$: 96 percent of the excess variance is the sample-specific term and 4 percent is the growing one. The sample length at which the growing term would matter as much as the fixed one is about 1608 years. That is why rescaling works, and it is the honest reason: on any sample a researcher will ever have, this channel is a fixed variance inflation rather than a divergence.

What survives. Three things, and they are what the paper claims. A correction applied with the *nominal* null does not address this channel at any family size — with fifty candidate signals, Bonferroni still gives 1.86 expected false positives and a 85 percent chance of at least one, against the 5 percent it promises, and even with ten candidates the chance is 49 percent. A correction applied with a *measured* null does address it, but the measurement is the hard part: the inflation is a property of the candidate population’s exposure distribution, so it has to be estimated on the population being screened rather than borrowed from this one. And a residual component does grow with the sample, so no fixed threshold contains it forever, though on realistic samples that component is small.

4.9 Dependence, and why the effective test count is the wrong summary

Dependence is measured with the marginals held exactly fixed. A block sign-flip permutation applied to the whole cross-section at once gives the null distribution of $\max_k |t_k|$ with dependence intact; the same permutation with an independent sign vector per strategy gives the same marginals with the columns independent. On 14 535 complete-history strategies the 95th percentile of $\max_k |t_k|$ is 4.39 jointly and 4.73 independently: dependence lowers the threshold by 7.2 percent.

Table 7: Five thresholds for the same family of tests, on one scale. *Nominal* is 1.96. *Calibrated* is the 95th percentile of the measured null for a single test. *Bonferroni* applies the nominal null with $M = 14,535$, the number of strategies with a complete 720-month history; the fourth ticker letter position exists only for the most recent 600 months, which is why that count is three quarters of the 19,380 and not all of them. The two max- t columns are the 95th percentile of the null maximum with the measured marginals, without and with cross-strategy dependence; the second is correct. *Bonf. error* is how much Bonferroni overshoots it. The two alpha rows mix two constructions and should be read as such: *Null SD* and *Calibrated* come from the marginal cross-sectional spread, which Section 4.4 shows is not a null, while the max- t columns come from the permutation that imposes $\alpha = 0$ given the exposures. Only the latter is used as a threshold.

Statistic	Wt.	Null SD	Nominal	Calibrated	Bonferroni	Max- t indep.	Max- t joint	Bonf. error
Mean return t	EW	0.889	1.96	1.77	4.64	4.73	4.39	0.057
Five-factor alpha t	EW	1.150	1.96	2.27	4.64	4.35	4.12	0.127
Mean return t	VW	1.026	1.96	2.08	4.64	4.57	4.31	0.078
Five-factor alpha t	VW	1.405	1.96	3.04	4.64	4.33	4.09	0.134

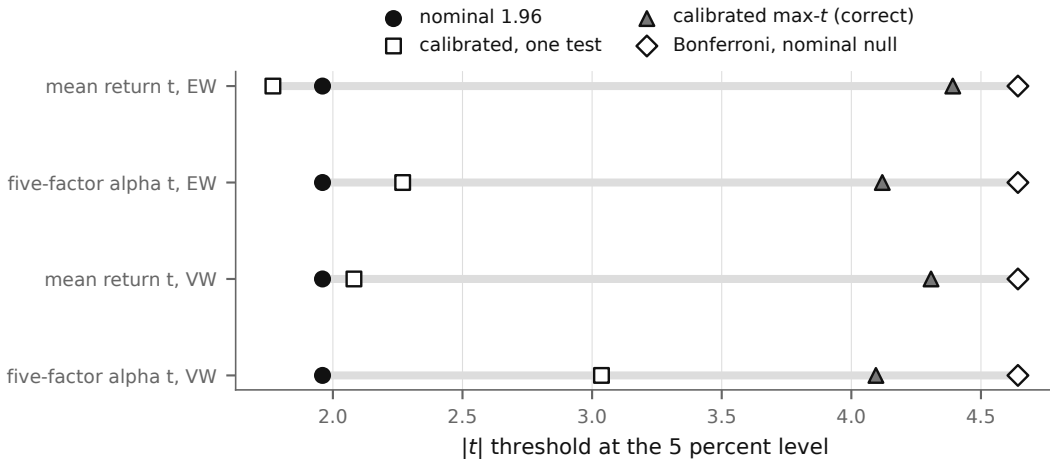


Figure 4: Five thresholds for one family of tests. Bonferroni with the raw trial count lands 5.7 percent above the correct calibrated maximum for the equal-weighted mean-return statistic and 13.4 percent above it for the value-weighted five-factor alpha. The uncorrected 1.96 is more than a factor of two below the correct threshold in every row.

Converting that into an effective number of independent tests is where the literature’s usual summary breaks down. On the same correlation matrix, Cheverud [2001] and Nyholt [2004] give 14 088 and Li and Ji [2005] gives 57, out of 14 535 — a disagreement of two orders of magnitude. The Šidák-implied count from the permutation maximum sits between them. The instability is not a defect of any one estimator: N_{eff} is an exponentially amplified rescaling of a threshold, so the 7.2 percent move measured above is a three- to five-fold move in the count. We therefore report the maximum-statistic quantile as the primary object and the effective count as a derived quantity, and we recommend that others do the same.

5 Thresholds, and the published predictors

Table 7 puts five thresholds for the same family of tests on one scale, and Figure 4 shows them.

Bonferroni with the raw trial count comes within 5.7 percent of the correct calibrated maximum for the mean-return t -statistic, and it overshoots by 13.4 percent for the five-factor alpha. It overshoots on both, and for two reasons that compound rather than cancel: it ignores dependence, which would lower the correct threshold, and it assumes a standard normal marginal that is *wider* and heavier-tailed than the measured one for the mean-return statistic — SD 0.889 and kurtosis 2.81 equal-weighted — which would lower it again. The margin is small on the mean return only because both effects are small there. The practical reading is that the conventional correction is close to right for

Table 8: Robustness suite. Every entry is reported whatever it shows. For the joint-versus-independent rows a value below one means the correct scheme gives a smaller null maximum, which is the whole point; for the expanding-window rows one is perfect and the nominal null’s own errors are in Section 6; for the exclusion and second-population rows the comparison is with the full-population value in Table 2. The individual numbers behind every row are in `data/results/robustness.json`.

Check	Quantity	Value
Joint vs independent resampling, finance	$\max\text{-} t $ 95th pct ratio	0.920
Joint vs independent resampling, cora	σ_{Δ} ratio	0.874
Joint vs independent resampling, citeseer	σ_{Δ} ratio	0.796
Joint vs independent resampling, pubmed	σ_{Δ} ratio	0.799
Joint vs independent resampling, ogbn-arxiv	σ_{Δ} ratio	0.820
Block length 1 to 12	spread in $\max\text{-} t $ 95th pct (%)	4.322
Expanding window, EW, cutoff 1.96	observed / predicted, calibrated	0.625
Expanding window, EW, cutoff 2.5	observed / predicted, calibrated	0.624
Expanding window, EW, cutoff 3.0	observed / predicted, calibrated	0.361
Expanding window, VW, cutoff 1.96	observed / predicted, calibrated	1.898
Expanding window, VW, cutoff 2.5	observed / predicted, calibrated	2.160
Expanding window, VW, cutoff 3.0	observed / predicted, calibrated	1.058
Second null population, EW	SD of mean-return t	0.662
Second null population, VW	SD of mean-return t	0.779
Drop alphabet-extreme sorts, EW	SD of mean-return t	0.912
Drop alphabet-extreme sorts, VW	SD of mean-return t	1.037
Balanced accuracy, cora	σ_{Δ} ratio to accuracy	0.994
Balanced accuracy, citeseer	σ_{Δ} ratio to accuracy	0.912
Balanced accuracy, pubmed	σ_{Δ} ratio to accuracy	0.896
Balanced accuracy, ogbn-arxiv	σ_{Δ} ratio to accuracy	0.698

the raw mean return and noticeably too strict for the factor-model alpha, and that in neither case does it get there by the route its user would describe.

The published predictors, briefly. Re-evaluating the 208 published predictors on their own original windows, 88 percent clear 1.96 uncorrected, Bonferroni leaves 83 and Benjamini–Hochberg 181; calibrating the null on the ticker population makes *more* of them survive, 102 and 190. We report this in one paragraph rather than as a finding, for two reasons that both point the same way. The measured null is narrow largely because of the zero-sum constraint of §4 — within a letter position every group appears equally often long and short — and a published predictor is under no such constraint, so transporting this null to that population is exactly the mistake §7 warns against. And Chen and Dim [2025] already conclude that this literature’s multiple-testing methods are too strict, so the direction is a numerical confirmation of theirs rather than a new result. The full table, the dependence-aware procedures and the Harvey and Liu [2015] haircuts are in Appendix A.

One caveat applies to every calibrated threshold in this paper. With 19,380 strategies the empirical tail resolves no finer than about 5×10^{-5} , so a calibrated cutoff past roughly $M = 200$ is a Gaussian extrapolation from the measured null rather than a measured quantile. We report it because the extrapolation is the same one Bonferroni makes and the comparison is the point, but it should not be quoted as a measurement.

6 Robustness

Six checks, with the underlying numbers in `data/results/robustness.json`.

Expanding-window calibration, and the one that partly fails. Read this one first, because it bounds what the threshold results below are worth. We fit the null on months before 1995 and ask how well it predicts exceedance rates after 1995. The scale is not stable across the two halves: it falls from 0.96 to 0.86 equal-weighted and rises from 0.91 to 1.02 value-weighted. Measured as mean absolute log error in the predicted exceedance rate across cutoffs 1.96, 2.5 and 3.0, the calibrated null scores 0.65 against the nominal null's 0.92 equal-weighted, and 0.49 against 0.39 value-weighted. So calibrating helps out of sample in one weighting and hurts in the other, and value-weighted it hurts substantially: the calibration under-predicts the observed exceedance rate by up to a factor of 2.2 across the three cutoffs. We take this as a real limit on how far a single unconditional calibration can be pushed, consistent with the decade-by-decade spread in §4. What it bounds is the *threshold* result of §5, which is the part that requires the calibration to transfer. It does not bound the mechanism of §4.4, which is an algebraic identity plus a measured exposure and needs no calibration to hold in any period.

The other five. *Alternative population:* the 210 higher-moment past-return strategies at horizons with no documented effect give a second, weaker null, narrower than the ticker population in level on both statistics but widened by risk adjustment in the same proportion (§4.6), which is the check that identified the alpha result as a general exposure mechanism rather than a quirk of ticker sorts. *Block length and replicate count:* the calibrated maximum moves by 4.3 percent across block lengths from 1 to 12, and is stable to within a percent from 8,000 replicates upward. *Joint against independent resampling:* quantified, and it is the difference reported in §4.9. *Alphabet-extreme exclusion:* leaves the calibration unchanged. *Subperiods and volatility regimes:* reported per decade and per regime.

7 What to do instead

Three recommendations follow from §4.4, in increasing order of how much work they cost.

Screen on the statistic that answers the question. If the question is whether a signal earns money, the raw mean return answers it and the alpha does not. On the known-null population $\Pr(|t_{\bar{r}}| > 3)$ is 0.44 percent against the nominal 0.27 percent, while $\Pr(|t_{\alpha}| > 3)$ is 5.2 percent. The alpha remains the right statistic for asking whether a strategy beats a benchmark, which is a different question and one where the exposure term is the answer rather than a contaminant.

Calibrate the cutoff on your own candidate population. On our population a value-weighted five-factor alpha needs 3.04 rather than 1.96 for a 5 percent per-test false positive rate, and the raw mean return needs 2.07; Table 3 has both for every model we run. We are deliberately not recommending 3.04 as a number to adopt, because this paper contains the evidence that it does not transfer: the second no-content population has a different alpha dispersion, and the cutoff follows the exposure distribution rather than the statistic. The recommendation is the procedure — form the population of signals you are willing to have searched, measure the spread of the statistic you intend to screen on over it, and take the quantile you want.

Two things make our number a floor rather than an estimate. A ticker letter induces factor exposure *accidentally*, whereas a candidate signal built from firm characteristics is related to size, value and profitability by construction, so a realistic search space should carry more exposure than this one and need a higher cutoff. And the tail resolves no finer than 5×10^{-5} on 19,380 strategies, so any cutoff quoted for a family much past $M = 200$ is an extrapolation.

Report both statistics. A signal whose evidence lives entirely in the exposure channel clears the alpha bar and not the raw-return bar, and the two counts together identify it where neither alone does. Recomputing the 208 published predictors on their own original windows, the two statistics correlate at 0.84; at the measured cutoffs 5 clear the alpha bar without clearing the raw-return bar, and 69 clear the raw-return bar without clearing the alpha one. At a common $|t| > 3$ on both the alpha-only count is 21. We report counts and distributions and name no predictor: the point is that the diagnostic is cheap, and on this population it mostly returns good news.

8 Other instances of the same failure

The finance case is the one we can measure, not the only one. The pattern is general: a search screens on a statistic S , the hypothesis of interest is “this candidate has no content”, and $E[S]$ under that hypothesis is not the value the threshold is set against. Three further instances, none of which needs the machinery of this paper to recognise once stated.

Accuracy against a uniform baseline. A classifier with no skill on a k -class problem is routinely compared against $1/k$. But a no-skill classifier that predicts the majority class scores the majority-class frequency, which on imbalanced data is far above $1/k$; the no-content value of the statistic is the base rate, not chance. The literature knows this and answers it exactly as data mining answered confidence, with re-centred statistics — balanced accuracy, Cohen’s κ , informedness — rather than with stricter significance thresholds. The parallel to alpha is close: the majority-class rate plays the role of $\hat{\beta}'\bar{f}$, a quantity that has nothing to do with the candidate’s content and everything to do with the population it is evaluated on.

Recommendation metrics with sampled negatives. Hit rate and NDCG computed against a sample of negatives rather than the full catalogue do not merely add noise; Krichene and Rendle [2020] show they are not consistent estimates of their exact counterparts, and that the ordering they induce over recommenders can differ from the true one. The no-content value again depends on a design choice — the number of sampled negatives — rather than on the recommender, so a threshold calibrated for one sampling rate is wrong for another.

Improvement over an estimated baseline. Any score of the form $\theta(\text{method}) - \theta(\hat{B})$ where the baseline $\hat{B}$ is itself selected or tuned inherits a non-zero expectation under the no-content hypothesis, because $\hat{B}$ is a maximum over its own search and is biased upward or downward depending on which side of the comparison did more searching. This is the same object that Hansen [2005] recentres in the Reality Check setting, and it is why the recentring step matters more than the multiplicity arithmetic that surrounds it.

What the instances have in common. In each case the correct response is to re-centre or re-calibrate the statistic on the population being searched, and in each case a multiplicity correction applied to the uncentred statistic will fail no matter how large the assumed trial count, because it is controlling the wrong error. Section 4.8 is what that failure looks like when it can be measured exactly. The general lesson, and the one we would want carried to a new domain, is that the first question about a screening statistic is not how many times it was computed but what it equals when there is nothing there.

9 The package

bzoo implements the twelve corrections of Table 9 behind one interface, so that a candidate population, a statistic and an assumed trial count produce a threshold without the caller reimplementing a procedure from a paper. It exists because the argument of this paper is that the null is the hard part and the correction is not, which is only a useful thing to say if the corrections are available to apply.

Two design decisions follow from the argument. There is no default null: every entry point requires the caller to supply the null distribution or the population to estimate it from, and raises rather than assuming a standard normal, because supplying that default is exactly the mistake §4.8 measures. And every correction has a unit test that reproduces either a published worked example or a published property of the procedure — that the three Hansen [2005] p -values are correctly ordered, that adding inferior candidates damages the Reality Check more than the test for superior predictive ability, that Benjamini–Yekutieli is more conservative than Benjamini–Hochberg by exactly the harmonic factor. There are 205 tests in continuous integration on Python 3.9 and 3.11.

10 Limitations

Several of these bound what our numbers mean.

Table 9: Corrections implemented in the package, with the source each one is tested against. Every method has a unit test that reproduces a published worked example or a published property of the procedure.

Method	Controls	Dependence	Source
bonferroni	FWER	none (independence assumed)	Bonferroni (1936)
sidak	FWER	none (exact under independence)	Sidak (1967), JASA 62, 626-633
holm	FWER step-down	arbitrary	Holm (1979), Scand. J. Statist. 6, 65-70
benjamini hochberg	FDR step-up	independence or positive regression dependence	Benjamini and Hochberg (1995), JRSS-B 57, 289-300
benjamini yekutieli	FDR step-up	arbitrary	Benjamini and Yekutieli (2001), Ann. Statist. 29, 1165-1188
storey qvalues	pFDR	weak	Storey (2002), JRSS-B 64, 479-498; Storey, Taylor and Siegmund (2004)
white reality check	FWER bootstrap, single composite hypothesis	learned from the re-sampling	White (2000), Econometrica 68, 1097-1126
hansen spa	FWER bootstrap, studentised and recentred	learned from the re-sampling	Hansen (2005), JBES 23, 365-380
romano wolf	FWER step-down bootstrap	learned from the re-sampling	Romano and Wolf (2005), Econometrica 73, 1237-1282
westfall young	FWER permutation, max-T and min-P	native	Westfall and Young (1993), Resampling-Based Multiple Testing
harvey liu zhu	haircuts on a reported Sharpe ratio	partial	Harvey, Liu and Zhu (2016), RFS 29, 5-68; Harvey and Liu (2015), JPM 42(1)
deflation	single-trial, via N and sigma	through the effective N	Bailey and Lopez de Prado (2014), JPM 40(5), 94-107

1. **The known-null population is null in raw returns, not in factor-model alphas** (§4.4). Two nulls follow from this and neither is guaranteed by construction the way the raw-return null is: the $\alpha = 0$ permutation used for testing rests on the sign-flip being valid given the exposures, and the marginal spread used for screening is a property of *this* population's exposures and does not transfer to another. The width of the alpha distribution is a property of the exposures a sort happens to induce, so its level is population-specific even though the widening replicates across our two populations. The number to quote is the mechanism, not the standard deviation.
2. **The mined population is a poor model of researcher behaviour.** A ticker-sorted portfolio is not what an economist tries. We claim only that these populations give the right *dependence structure and tail behaviour* for the corrections to consume, and, for the mechanism, the right kind of accidental factor exposure. The corroboration we can offer is narrower than we would like: the calibration is reproduced on the four disjoint letter-position subsets of the ticker population and on the higher-moment past-return strategies at horizons with no documented effect (Table 5), which is two independent constructions rather than the several the design would ideally have. The 114 documented placebo characteristics would be the right third, and building their portfolios from the distributed firm-level signals is work we have not done.
3. **Trial counts are lower bounds.** Unpublished attempts are invisible. Every threshold is reported over five orders of magnitude of N for this reason.

4. **Calibration precision, and drift.** On roughly 4,845 dependent strategies the null standard deviation is pinned only to the range the four disjoint letter-position subsets span — 0.739 to 1.045 on the mean-return statistic equal-weighted, and 1.103 to 1.590 on the five-factor alpha value-weighted — and not to two decimal places. It also drifts across the sample, and an expanding-window check shows a calibration fitted on the first half transfers imperfectly to the second — helping in one weighting and hurting in the other (§6). A conditional calibration is the right object and we have not built one.
5. **One factor-model family.** Everything here uses the Fama–French line, from the CAPM to the five-factor model plus momentum. We do not run the q -factor model or a Barillas–Shanken specification, so whether the inflation is of the same size against a different benchmark is open. The identity is invariant to the choice, so the mechanism is not in doubt; its magnitude is.
6. **One market, one period, one provider.** US equities, 1963 to 2022, with the strategies as Chen and Dim [2025] built them and the factors as Kenneth French publishes them. The construction that gives us ground truth also ties us to one vendor’s universe.
7. **Tail resolution limits every cutoff we quote.** With 19,380 strategies the empirical tail resolves no finer than 5×10^{-5} , so any calibrated threshold past about $M = 200$ is a Gaussian extrapolation rather than a measured quantile (§5).
8. **The exposure channel is not identified.** We show the alphabet maps onto profitability and investment loadings, and we cannot say why (§4.6). Industry composition is the obvious candidate, and testing it needs firm-level holdings the distributed portfolio returns do not contain.

11 Broader impact

This work re-evaluates a published record, and the finding runs against the direction that would be comfortable: on the statistic the field screens on, the conventional bar is too *lenient*, and content-free strategies clear $|t_\alpha| > 3$ at 19 times the nominal rate. We report counts and distributions and name no paper as a false discovery, both because the diagnostic is meant to be run by authors on their own candidates and because running the diagnostic on the published set finds most of the evidence surviving on both statistics. The foreseeable misuse is quoting one of our cutoffs as a threshold in a setting whose null has not been measured, which is the error the paper exists to describe; the package raises rather than supplying a default.

12 Availability and reproducibility

`make all` reproduces every table and figure in this paper from the raw sources with one command. Every number in the text is a macro emitted by the pipeline; none is typed by hand. The datasheet describing every released file is in `DATASHEET.md`, and the decision log in `DECISIONS.md` records each non-obvious choice on the day it was made, including the ones we got wrong first and what the corrected version changed. We redistribute derived statistics only, never the underlying returns, which remain the authors’ to distribute.

13 Conclusion

A search over a large candidate space needs a null for the statistic it screens on, and correcting for the number of trials does nothing if that statistic is non-zero when there is no content. Data mining learned this from association rules and answered it by re-centring the statistic rather than raising the threshold. The same failure is present in accuracy against a uniform baseline, in recommendation metrics with sampled negatives, in any improvement-over-an-estimated-baseline score, and — measurably, because there the ground truth is known — in the factor-model alpha that empirical asset pricing screens candidate signals on.

On 19,380 strategies that are null by construction the raw mean-return t -statistic sits where the textbook says, at 1.026, and the five-factor alpha t -statistic does not, at 1.405, with 5.2 percent of empty strategies clearing $|t| > 3$. Least squares says why exactly. Four checks put the channel on

Table 10: The 212 published predictors of Chen and Zimmermann [Chen and Zimmermann, 2022], re-evaluated on each predictor’s own original sample window. 208 have at least 60 months. Calibrating the null makes *more* of them survive, because the measured null for the equal-weighted mean-return t -statistic is narrower than the nominal one, not wider. Two rows need a note. The Storey q -value keeps all 208 under either null because the estimated null proportion $\hat{\pi}_0$ is essentially zero on this population, which is what a procedure designed for a mostly-null family does when handed a family of published claims; it is not a bug. The three Harvey–Liu–Zhu rows use the $M = 1,000$ their own procedure assumes rather than the 208 of the rows above, so their survival counts are not comparable with the rest of the column.

Correction	Controls	M	Survive, nominal null	Survive, calibrated null	Change
Uncorrected	per test	208	183	191	8
Bonferroni	FWER	208	83	102	19
Sidak	FWER	208	85	102	17
Holm	FWER	208	90	110	20
Benjamini-Hochberg	FDR	208	181	190	9
Benjamini-Yekutieli	FDR	208	134	152	18
Storey q -value	pFDR	208	208	208	0
Harvey-Liu-Zhu (bonferroni)	FWER	1 000	71	86	15
Harvey-Liu-Zhu (holm)	FWER	1 000	71	88	17
Harvey-Liu-Zhu (bhy)	FDR	1 000	94	119	25

Table 11: Survival as a function of the assumed total number of tests M , which is never fixed to one value. The smallest M is the number of published predictors themselves, which is the only M that is observed rather than assumed.

Null	M	t threshold	Bonferroni	Holm	BHY
nominal	208	3.67	83	90	134
nominal	1 000	4.06	71	71	94
nominal	10 000	4.56	57	57	70
nominal	100 000	5.03	47	47	54
calibrated	208	3.27	102	110	152
calibrated	1 000	3.61	86	88	119
calibrated	10 000	4.06	71	71	86
calibrated	100 000	4.47	60	60	69

the exposures rather than the regression: the spread tracks which factors the sort loads on rather than how many regressors are added; across exposure deciles it runs from 0.994 to 1.810 with the numerator growing by 1.98 and the denominator by 1.19; a zero-exposure placebo returns 1.001; and a second, unrelated no-content population is widened by the same factor.

Run on that population, Bonferroni at $M = 19,380$ rejects 0 strategies on the mean return and 69 on the alpha, and substituting the measured null takes that to 0. So the multiplicity machinery is sound and the null it is given is not, and because 96 percent of the excess variance is a term that does not grow with the sample, a rescaling is enough — on any sample a researcher will have. What it is not is transferable: the rescaling is a property of the candidate population’s exposures, and since a ticker letter induces exposure accidentally while a signal built from firm characteristics is related to size, value and profitability by construction, our numbers are a floor rather than an estimate. The first question to ask about a screening statistic is not how many times it was computed, but what it equals when there is nothing there.

A The published predictors in full

Table 11 and Figure 5 report every threshold as a function of the assumed total number of tests M rather than at one value, because only the smallest M in the table is observed and the rest are assumptions. One caveat applies to every calibrated threshold in this section. With 19,380 strategies the empirical tail resolves no finer than about 5×10^{-5} , so a calibrated cutoff past roughly $M = 200$ is a Gaussian extrapolation from the measured null rather than a measured quantile. We report it

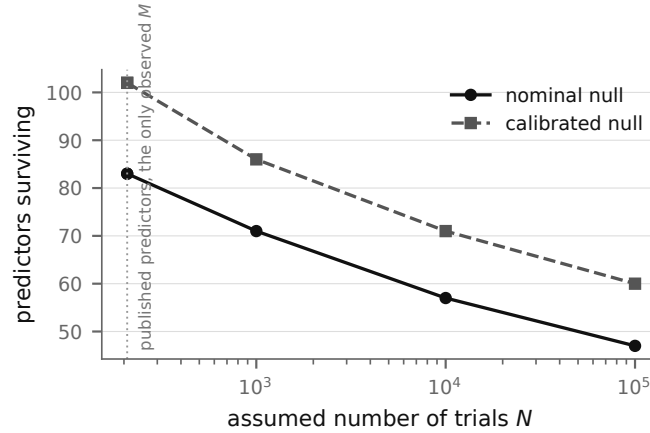


Figure 5: Survival of the published predictors as a function of the assumed number of tests, under the nominal and the calibrated null. The marked value is the number of published predictors, the only M that is observed.

because the extrapolation is the same one Bonferroni makes, and the comparison is the point; but it is an extrapolation and should not be quoted as a measurement.

The dependence-aware procedures agree with the reading in §5. On the common-sample panel the Reality Check gives $p = 0.0010$ and the test for superior predictive ability $p = 0.0005$, so the best of the published predictors comfortably beats zero after accounting for the search over all of them; the Romano and Wolf [2005] step-down procedure rejects 72 of the 183 predictors with a common sample window. The median Harvey and Liu [2015] haircut is 48 percent of the reported Sharpe ratio.

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